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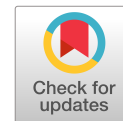
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Modeling Solute Transport in the WinSRFR Surface Irrigation Software

Eduardo Bautista, M.ASCE¹; and James L. Schlegel²

Abstract: A solute transport model has been added to the WinSRFR surface irrigation software package. The model uses advective-dispersive principles to predict the flow of solute with the irrigation stream and its final longitudinal distribution along the field based on water flow predictions generated by the simulation engine. Computations are based on the fractional stepping method: advection is computed using the method of characteristics from the velocity field computed by the simulation engine water flow calculations, and dispersion is calculated from the advection results using a centered-difference finite-difference scheme. The Akima interpolation scheme was used to build the method-of-characteristics solution. The computations are robust and execute in little time. The accuracy of the model was evaluated with numerical tests and by comparison with field measurements. Results suggest that the model can generate useful predictions under typical irrigation conditions. DOI: [10.1061/\(ASCE\)IR.1943-4774.0001507](https://doi.org/10.1061/(ASCE)IR.1943-4774.0001507). © 2020 Published by American Society of Civil Engineers.

Introduction

A common practice in surface irrigation is the application of nitrogen fertilizer with the irrigation stream. In principle, fertigation promotes high nutrient recovery by the crop and reduces the potential for nutrient leaching below the root zone because it facilitates the application closer to development stages with high nutrient demand. This is difficult to do with mechanical fertilizer application systems. However, effective fertigation practices require an irrigation system with excellent distribution uniformity and application efficiency. Moreover, even with an efficient water application, nutrient distribution along the length of the field and vertically within the soil profile can be poor if the timing of the nutrient injection relative to the flow of the irrigation stream is inadequate (Playan and Faci 1997). In furrow irrigation systems, where infiltration is two-dimensional, water and fertilizer distribution can be further affected by irrigation system management variables such as flow depth (Abbasi et al. 2003a) or the use of alternate furrow irrigation (Ebrahimian et al. 2013).

Because of the need to understand and manage the distribution of nutrients applied through surface irrigation systems, solute transport modeling tools have been under development for over 20 years. This development is closely linked to the development of tools for modeling surface irrigation flows, as well as tools for modeling the movement of water and solutes within the soil profile. The development discussed herein deals only with modeling the transport of solute with the surface and flow and its ultimate distribution along the length of the field.

The flow of fertilizer with the irrigation stream is a special case of advective-dispersive transport. Advection refers to the movement of solutes with the mean velocity of the conveying fluid, and thus as a result of the mean forces acting on that flow, whereas dispersion considers the transport and mixing of the solute as a result of velocity gradients within a cross section, flow turbulence, and molecular diffusion. Numerous studies have discussed the principles of advective-dispersive transport in open-channel water flows (e.g., Fischer et al. 1979; Cunge et al. 1980). The analysis assumes a neutrally buoyant solute and relatively small concentrations. These conditions imply that the solute has no effect on the density of the water and, consequently, on the irrigation flow. The analysis also assumes that the constituent is not subject to chemical reactions and that the only sink is the solute lost via infiltration. If the solute is applied in a pulse of short duration relative to the water flow event of interest, dispersion causes the concentration of the pulse to spread and attenuate as it travels downstream.

Solving the problem of solute advection-dispersion in surface irrigation also requires solving the unsteady surface-flow problem coupled to infiltration. Solution methods that have been developed for the combined problem can be grouped in three categories. The simplest but least accurate approach decouples the one-dimensional irrigation flow calculations from the solute transport, with the former executed first to provide the computational grid for the latter (Boldt et al. 1994; Playán and Faci 1997; Strelkoff et al. 2006). Mass balance is used to track the fate of discrete parcels of water, with or without solute, entering a field. The parcels are defined by the time step of the irrigation simulation. Water entering the stream during a time step pushes water already on the field further downstream and does not mix with it. Water is abstracted from the parcel by infiltration, with the calculations conducted using the cells of the irrigation simulation, cell by cell. This requires the water flow cells to be calculated with an Eulerian computational grid. The remaining surface water is water that can flow downstream. The parcel is tracked in subsequent time steps until all its water is exhausted. Because calculations allow the solute to be transported only with the bulk flow of water, they produce a reasonable approximation to advective transport.

The second solution method also decouples the water and solute flow calculations, but computes the latter with the partial

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differential equation of advection-dispersion (ADE) (García-Navarro et al. 2000; Abbasi et al. 2003b; Zerihun et al. 2005a; Perea et al. 2010). This approach allows using different computational procedures for each of the two flow components and, therefore, allows superimposing the solute transport calculations on an existing water flow model. As with the water flow equations, the ADE equation can only be solved numerically for practical problems. Solving the water and solute flow problems can fail to properly conserve the solute, however (Murillo et al. 2005). Furthermore, the ADE alone presents substantial computational challenges. Because pure advection is described by a hyperbolic partial differential equation whereas pure dispersion by a parabolic type equation, numerical solutions to the combined equation can introduce numerical dispersion larger than the predicted dispersion (Islam and Chaudhry 1997).

The fractional stepping method (also known as split operator) was developed to mitigate this problem. In fractional stepping, the advection component is computed first from the velocity field computed by the water flow simulation by using the method of characteristics to track the fate of particles introduced at the boundary. Dispersion is calculated subsequently from the concentration field computed by the advection solution. This splitting approach is subject to a numerical error that is proportional to the calculation time step (LeVeque 2002). Furthermore, because it requires many interpolations, it can fail to properly conserve mass. Thus, different variations of the method have been reported. Most decoupled fertigation models have used fractional stepping to deal with the ADE calculations (García-Navarro et al. 2000; Zerihun et al. 2005a; Perea et al. 2010).

A third solution strategy consists of computing the water flow and ADE equations simultaneously at every space and time step of the calculations. Currently, the preferred approach for solving the fully coupled problem is using finite-volume numerical procedures. This approach has been used to model one-dimensional surface flow cases with both borders and furrows (Burguete et al. 2009; Zhang et al. 2016; Brunetti et al. 2018), and two-dimensional border/basin irrigation (Dai et al. 2019). Finite-volume methods were originally developed to model compressible flows, in which shock waves can easily develop, and are therefore inherently better suited for handling discontinuities in the solution and boundary conditions than finite-difference methods. Because those methods are computationally expensive in comparison with existing finite-difference solutions, their application to modeling irrigation flows is a topic of active research.

Perea et al. (2010) reported a solute transport model based on the ADE and the fractional step solution method. This approach, with modifications, has been added to the simulation the component of WinSRFR version 5.1 (Bautista and Schlegel 2019), a software package for the analysis of surface irrigations systems developed by the USDA Agricultural Research Service. The decoupled solution takes advantage of the WinSRFR hydraulic solution, which runs prior to the advection-dispersion calculations. This article documents the model and examines its strengths and limitations.

Governing Equations

Water Flow

SRFR 5, the simulation engine of WinSRFR (Bautista et al. 2016), uses the zero-inertia form of the one-dimensional shallow-water (depth-averaged) unsteady open-channel flow equations (Strelkoff and Clemmens 2007) to model surface irrigation flows

$$\frac{\partial A_y}{\partial t} + \frac{\partial Q}{\partial x} + \frac{\partial A_z}{\partial t} = 0 \quad (1)$$

$$gA_y \left(\frac{\partial y}{\partial x} - S_0 + S_f \right) = 0 \quad (2)$$

Eq. (1) represents conservation of mass and states that the time t (T) rate of change in the surface volume per unit length A_y (L^3/L) is a function of the rate of change with distance x (L) of flow rate Q (L^3/T) and the time rate of change of water loss due to infiltration A_z (L^3/L). Eq. (2) represents conservation of linear momentum but without the local and convection acceleration terms. Thus, the gradient in hydrostatic pressure ($gA_y \partial y / \partial x$) force and the weight component of the water in the direction of flow ($gA_y S_0$) are in balance with the frictional resistance by the soil surface and vegetation ($gA_y S_f$). In this expression, g = acceleration of gravity (L/T^2); y = flow depth (L); S_0 = field bottom slope (L/L); and S_f = friction slope (L/L). The zero-inertia equations are reasonably accurate in comparison with the full hydrodynamic equations flow at low Froude numbers typical of irrigation flows (Strelkoff and Clemmens 2007). Because they can represent the motion of waves traveling both downstream and upstream, they can be applied to free-draining and blocked-end irrigation systems.

When the slope is uniform and relatively steep (for practical purposes, slopes greater than 0.004 m/m), the pressure gradient term in Eq. (2) becomes negligible because normal flow conditions are quickly attained. SRFR then replaces that equation with

$$S_f = S_0 \quad (3)$$

The resulting kinematic wave equations are accurate in comparison with zero inertia under the previously described conditions but can only be applied to free-draining systems because they can only represent a wave traveling downstream.

The irrigation flow equations represent a coupled surface–subsurface flow problem because of the infiltration term A_z . SRFR 5 offers several options for representing infiltration (Bautista 2016; Bautista and Schlegel 2019). Empirical equations assume that A_z is a function of opportunity time, i.e., the time of water contact with the soil surface. When dealing with furrows, where infiltration is a two-dimensional flow process, expressions are available for adjusting the average intake as a function of a nominal inflow-rate-dependent wetted perimeter. Infiltration can also be computed as a function of soil hydraulic properties, initial soil water content, and the variable water pressure at the soil surface, semiempirically, with the Green-Ampt equation (borders/basins) and the Warrick-Green-Ampt equation (furrows), or physically with the Richards equation (only borders). Evidently, selection of an infiltration modeling approach and parameterization of the selected equation play a key role in predicting the surface flow of water, the final distribution of infiltrated water, and consequently, the infiltration of solute along the field and vertically in the soil profile, but this critical issue will not be addressed here.

Initial conditions for the water flow equations are

$$y(x, 0) = 0, \quad Q(x, 0) = 0 \quad (4)$$

At the upstream boundary, the condition is the specified inflow hydrograph $Q_0(t)$

$$Q(0, t) = Q_0(t) \quad (5)$$

Downstream boundary conditions are needed only when solving the zero-inertia equations. During the advance phase, depth and flowrate are nil at the advancing front x_a

$$y(x_a, t) = 0; Q(x_a, t) = 0, \quad t \leq t_L \quad (6)$$

where t_L = final advance time. After water reaches the end of the field L_f , a depth-discharge relationship applies at the downstream boundary for a free draining system

$$y(L_f, t) = 0, \quad t > t_L \quad (7)$$

For a blocked system, the discharge is zero

$$Q(L_f, t) = 0, \quad t > t_L \quad (8)$$

Solute Flow

The conservative, depth-averaged ADE equation for one-dimensional flow of a nonreactive solute in the irrigation stream is

$$\frac{\partial(A_y C)}{\partial t} + \frac{\partial(A_y v C)}{\partial x} + C \frac{\partial A_z}{\partial t} = \frac{\partial(A_y K_x \frac{\partial C}{\partial x})}{\partial x} \quad (9)$$

where C = concentration (M/L^3); v = cross-sectional averaged flow velocity (L/T); K_x = coefficient of longitudinal dispersion (L^2/T); and A_y = surface flow area, as previously defined. This equation states that the time rate of change of solute mass is a function of the rate of change solute flux with distance, the solute lost to infiltration, and solute dispersion (Cunge et al. 1980). The dispersion coefficient represents the effect of velocity variations in the transverse and vertical directions with respect to the cross-sectional averaged velocity as well as the effect of turbulent mixing within a cross section, which attenuates the effect of velocity variations. Thus, it is the mechanism by which the three-dimensional transport process is simplified to one dimension. Use of a Fick-type law to represent dispersion [right-hand side of Eq. (9)] assumes that the solute is fully mixed within the cross section, a condition that is likely to be satisfied in irrigated fields.

Eq. (9) can be simplified by expanding to

$$\frac{A_y \partial C}{\partial t} + \frac{C \partial A_y}{\partial t} + \frac{A_y v \partial C}{\partial x} + \frac{C \partial A_y v}{\partial x} + C \frac{\partial A_z}{\partial t} = \frac{\partial(A_y K_x \frac{\partial C}{\partial x})}{\partial x} \quad (10)$$

and using Eq. (1) to represent the infiltration sink term and substituting $Q = AV$. The resulting nonconservative ADE is the equation that has been used by previous authors (García-Navarro et al. 2000; Zerihun et al. 2005a; Perea et al. 2010)

$$\frac{A_y \partial C}{\partial t} + \frac{Q \partial C}{\partial x} = \frac{\partial(A_y K_x \frac{\partial C}{\partial x})}{\partial x} \quad (11)$$

with initial conditions given by

$$C(x, 0) = 0 \quad (12)$$

and the upper boundary conditions specified as follows:

$$C(0, t) = C_0(t) \quad (13)$$

where the inflow concentration $C_0(t)$ is a function of solute solution injection rate, concentration in a supply tank, and water discharge to the field. Downstream boundary conditions will be discussed subsequently.

Numerical Solution

Zero-Inertia/Kinematic Wave Equations

Following Perea et al. (2010), the solution used herein decouples the water and solute transport equations. Solution of the water flow equations have been discussed by Walker and Skogerboe (1987), Strelkoff (1990), Strelkoff and Clemmens (2007), and Bautista et al. (2016). Details of those calculations will not be discussed here, but it is relevant to this presentation to review the discretization approach used by the SRFR simulation engine (i.e., the SRFR dynamic-link library). SRFR solves the flow equations using an Eulerian finite-difference scheme, with the spatial increment initially specified from the user-editable cell density parameter. During the advance phase, the solution marches in time by advancing the stream to the next specified grid distance and, therefore, solves for the time step. The space increments are not constant, however, depending on spatial variations in field properties (e.g., field elevation). Furthermore, the space step is reduced whenever the stream experiences substantial deceleration. In addition, the code switches to solving for the advance distance with the time increment known to enforce user-specified boundary conditions, for example, changes in the inflow rate or solute injections.

During postadvance calculations, when the spatial grid is fully defined, the solution can advance in principle using constant time steps. However, as with the space steps, the time step is adjusted depending on boundary conditions and computational situations, for example, failure of the solution to converge. As a result, the computational grid is irregular. The significance of this issue is that numerical solutions to the ADE equation, such as the one discussed subsequently, generally assume a uniform computational grid.

Advection Equation

The fractional stepping approach (LeVeque 2002) solves at each time step the pure advection problem first

$$\frac{A_y \partial C}{\partial t} + \frac{Q \partial C}{\partial x} = 0 \quad (14)$$

Using the advected values of C , the pure dispersion problem is solved next

$$\frac{A_y \partial C}{\partial t} = \frac{\partial(A_y K_x \frac{\partial C}{\partial x})}{\partial x} \quad (15)$$

The solution to Eq. (14) is obtained by writing that expression as follow:

$$\frac{\partial C}{\partial t} + \frac{v \partial C}{\partial x} = 0 \quad (16)$$

By the chain rule of differentiation, this is the total derived of C with respect to time, hence

$$\frac{dC}{dt} = 0 \quad (17)$$

along the characteristics curve given by

$$\frac{dx}{dt} = v(x, t) \quad (18)$$

The solution to the advective problem consists, then, of tracing the characteristic curves along which concentrations specified with the boundary conditions travel unchanged.

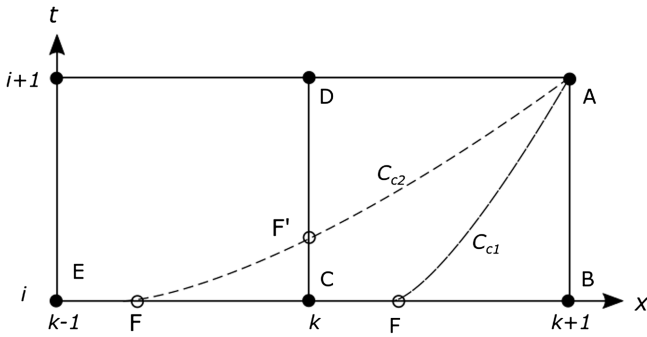


Fig. 1. Calculation of a characteristic segment from the water solution computational grid.

Fig. 1 shows the advection numerical solution. The figure represents two cells of the water flow solution, which completes before initiating the ADE calculations. Water flow variables, including the cross-sectional average velocity, are known at each point (i, k) of the rectangular grid and are available in computer memory. At time i , concentration values are known at the spatial grid points $\dots, k-1, k, k+1, \dots$. A solution is sought at the grid point $i+1, k+1$ (Node A). The concentration at Node A depends on the concentration at the point F, which is the foot of the characteristic at the time i . Two possible characteristic segments are shown in the figure. The one labeled C_{c1} is entirely contained within a grid cell, and the one labeled C_{c2} spans two cells. Calculation of the concentration at F involves the following steps (Perea et al. 2010).

Step 1: Calculation of Characteristic Segments Backward from Time $i+1$ to i

A first-order finite-difference approximation to Eq. (18) is

$$\frac{x_A - x_F}{\Delta t_n} = \frac{(v_F + v_A)}{2} \quad (19)$$

This provides an expression for the location of the foot of the characteristic x_F

$$x_F = x_A - \frac{(v_F + v_A)}{2} \Delta t_n \quad (20)$$

If the unknown velocity at F is expressed as a weighted average of the velocities at nodes k and $k+1$, then

$$\phi = \frac{(x_F - x_C)}{(x_A - x_C)} \quad (21)$$

$$v_F = \phi v_B + (1 - \phi) v_C \quad (22)$$

Then, x_F and v_F can be determined by iterating through Eqs. (20)–(22) until satisfying the convergence criterion

$$|v_F^{j+1} - v_F^j| < \text{tol} \quad (23)$$

where j = iteration index; and tol = tolerance value. The procedure is initialized by setting $v_F = v_C$. This is all that is needed when F is the foot of C_{c1} .

When dealing with a characteristics segment such as C_{c2} , the calculations will force x_F beyond x_k (Node C). The procedure then calculates the time $t_{F'}$ at which the characteristic crosses x_k , and continues the search for the foot of the characteristic. Equations like the ones used to located x_F are used to find $t_{F'}$

$$t_{F'} = t_A - \frac{2(x_A - x_C)}{(v_A + v_{F'})} \quad (24)$$

$$\psi = \frac{(t_{F'} - t_C)}{(t_D - t_C)} \quad (25)$$

and

$$v_{F'} = \psi v_D + (1 - \psi) v_C \quad (26)$$

This alternate calculation is initialized with v_E . The convergence criterion is as in Eq. (23).

After solving for $t_{F'}$, the search for x_F can continue using Eqs. (20)–(22) but with Nodes A, B, C, and D reassigned to their new locations (e.g., Node F' becomes the new Node A). If the solution forces x_F beyond x_{k-1} , then the characteristic segment spans three or more cells. Eqs (24)–(26) need to be used again to find a time $t_{F'}$ at which the segment crosses x_{k-1} , and after which the search continues. Evidently, if k is the upstream boundary node, $t_{F'}$ is the foot of the characteristic and with the concentration given by the boundary condition.

Step 2: Interpolation of the Concentration at x_F from Known Concentrations at Time i

The difficulty in computing the concentration at x_F is that solute concentration changes at the boundary will typically occur as step changes, which are difficult to accurately capture from changes in conditions at neighboring computational nodes. Different interpolation schemes have been tested to deal with this aspect of the method-of-characteristics solution to the advection equation. Perea et al. (2010) used a cubic spline, but tests conducted with this method proved unsatisfactory under some conditions. Better results have generally been obtained with Akima interpolation (Akima 1970).

Akima interpolation is a cubic spline scheme where the interpolating function for the interval $[x_k, x_{k+1}]$ is given by

$$f(x) = a_0 + a_1(x - x_i) + a_2(x - x_i)^2 + a_3(x - x_i)^3 \quad (27)$$

The coefficients of Eq. (27) are computed subject to

$$\begin{aligned} f(x_k) &= \zeta_k \\ f(x_{k+1}) &= \zeta_{k+1} \\ f'(x_k) &= \hat{s}_k \\ f'(x_{k+1}) &= \hat{s}_{k+1} \end{aligned} \quad (28)$$

where ζ = known value of $f(x)$ (in this case, the concentration values); and $\hat{s}$ = weighted average slope of the curve defined by the discrete points ζ_k , computed as follows:

$$\hat{s}_k = \frac{|s_{k+1} + s_k|s_{k-1} + |s_{k-1} + s_{k-2}|s_k}{|s_{k+1} + s_k| + |s_{k-1} + s_{k-2}|} \quad (29)$$

where s = slopes of individual segments, given by

$$s_k = \frac{(\zeta_{k+1} - \zeta_k)}{(\zeta_{k+1} - \zeta_k)} \quad (30)$$

An important difference between the conventional cubic and the Akima spline is that the former assumes continuous second derivatives whereas the second does not. Thus, the Akima interpolation reduces artificial oscillations induced by rapid changes in the estimated second derivative.

Step 3: Calculation of Concentration at t_{i+1} from the Known Concentrations

This value is either the interpolated value at time i or the upstream boundary condition.

In blocked-end irrigation systems, velocities can be zero or near-zero after cutoff, meaning that the characteristic curves are near vertical. Under these conditions, advection calculations stop and the concentration at Node A is simply assumed to be equal to that of Node B.

Dispersion Equation

The Crank-Nicolson method has proven an effective method for solving the dispersion problem in previous studies (Zerihun et al. 2005a; Perea et al. 2010). It is a centered-difference, second-order-accurate scheme applicable to parabolic differential equations (LeVeque 2002). Application of the scheme to a grid point k yields

$$\frac{C_k^{i+1} - C_k^*}{\Delta t_i} = \frac{\theta}{A_{yi}(x_{k+1} - x_{k-1})} \left[A_y K_x \frac{\partial C}{\partial x} \Big|_{k+1/2}^{i+1} - A_y K_x \frac{\partial C}{\partial x} \Big|_{k-1/2}^{i+1} \right] + \frac{1-\theta}{A_{yi}(x_{k+1} - x_{k-1})} \left[A_y K_x \frac{\partial C}{\partial x} \Big|_{k+1/2}^* - A_y K_x \frac{\partial C}{\partial x} \Big|_{k-1/2}^* \right] \quad (31)$$

where θ = dimensionless time weighting factor, set to 0.5; and C^* = advection solution. Discretization of the spatial derivative and grouping of common terms produces

$$\alpha_k C_{k-1}^{i+1} + \beta_k C_k^{i+1} + \gamma_k C_{k+1}^{i+1} = \delta_k \quad (32)$$

The Appendix explains the terms α , β , γ , and δ , and additional details have been given by Perea et al. (2010) and Zerihun et al. (2005a).

Application of Eq. (32) to all interior nodes produces a tridiagonal system of linear equations. Press et al. (2007) described an efficient solution algorithm for tridiagonal systems of equations, which, furthermore, allow one to store the coefficients α_k , β_k , γ_k , and δ_k in separate vectors. A goal in the implementation of the boundary conditions is the preservation of the tridiagonal structure. In contrast with the advection solution, upstream and downstream boundary conditions are needed to solve the dispersion equation.

For Node 0, the upstream boundary condition prior to cutoff is the prescribed concentration Eq. (13). For Node 1, and because the concentration at Node 0 is given, Eq. (32) is written

$$\beta_1 C_1^{i+1} - \gamma_1 C_2^{i+1} = \delta_1 - \alpha_1 C_0^{i+1} \quad (33)$$

During advance, no solute can flow beyond the leading edge of the stream. The downstream boundary condition is then a zero-concentration gradient

$$\frac{\partial C}{\partial x} = 0 \quad (34)$$

Perea et al. (2010) used a centered finite difference to approximate Eq. (34), which requires adding a ghost node to the calculations. A simpler approach was adopted herein by using second-order backward differences. The approximation (LeVeque 2007) was developed considering the possibility of cells of unequal size

$$\frac{\partial C}{\partial x} \approx \frac{(r+1)}{\Delta x_N r} C_N - \frac{r}{\Delta x_N (r-1)} C_{N-1} + \frac{1}{\Delta x_N (r-1)r} C_{N-2} \quad (35)$$

where

$$\Delta x_N = x_N - x_{N-1} \quad (36)$$

and

$$r = \frac{x_N - x_{N-2}}{x_N - x_{N-1}} \quad (37)$$

Thus, when the cells are of equal size, $r = 2$. Combining Eqs. (35) and (34) yields

$$C_N^{i+1} = \left(\frac{r^2}{r^2 - 1} \right) C_{N-1}^{i+1} + \left(\frac{1}{1 - r^2} \right) C_{N-2}^{i+1} = f_1 C_{N-1}^{i+1} + f_2 C_{N-2}^{i+1} \quad (38)$$

where the f coefficients on the right side are the terms in parentheses on the left. Combining with Eqs. (38) and (32) yields the expression for node $N - 1$

$$(\alpha_{N-1} + f_2 \gamma_{N-1}) C_{N-2}^{i+1} + (\beta_{N-1} + f_1 \gamma_{N-1}) C_{N-1}^{i+1} = \delta_{N-1} \quad (39)$$

Because Eq. (39) cannot be used during the first two time steps, a low-order finite-difference approximation is used instead, from which

$$C_N^{i+1} = C_{N-1}^{i+1} \quad (40)$$

Hence, for the first time step, $C_1^{i+1} = C_0^{i+1}$, and for the second

$$C_1^{i+1} = \frac{\gamma_1 - \alpha_1 C_0^{i+1}}{\beta_1 + \delta_1} \quad (41)$$

After the stream reaches the end of the field, Eq. (34) continues to apply if the downstream boundary is blocked but not with a free-draining irrigation system because the solute can flow beyond the end of the field. In that case, the concentration gradient of the diffusion solution is assumed equal to the advection flow gradient (Zerihun et al. 2005a; Perea et al. 2010)

$$\frac{\partial C}{\partial x} \Big|^{i+1} = \frac{\partial C^*}{\partial x} \Big|^{i+1} \quad (42)$$

Using Eq. (35) to represent Eq. (42) yields

$$C_N^{i+1} = C_N^* + f_1 (C_{N-1}^* - C_{N-1}) + f_2 (C_{N-2}^* - C_{N-2}^{i+1}) \quad (43)$$

in which f_1 and f_2 are as in Eq. (38). After combining with Eq. (33), the equation for node $N - 1$ with a free-draining boundary is

$$(\alpha_{N-1} + f_2 \gamma_{N-1}) C_{N-2}^{i+1} + (\beta_{N-1} + f_1 \gamma_{N-1}) C_{N-1}^{i+1} = \delta_{N-1} - \varepsilon_{N-1} \quad (44)$$

where

$$\varepsilon_{N-1} = \gamma_{N-1} (C_N^* - f_1 C_{N-1}^* - f_2 C_{N-2}^*) \quad (45)$$

After water is shut off, the upstream boundary condition is a solute zero-gradient, like the downstream condition. Using a second-order forward finite-difference approximation yields an expression for the concentration at the boundary

$$C_0^{i+1} = \left(\frac{r^2}{r^2 - 1}\right) C_1^{i+1} - \left(\frac{1}{r^2 - 1}\right) C_2^{i+1} = f_1 C_1^{i+1} - f_2 C_2^{i+1} \quad (46)$$

This notation assumes that the index of the upstream node is always zero and, thus, that node indices are adjusted as the stream contracts. In contrast with Eq. (37), r is now the ratio of the length of the first two cells to the first. Eq. (46) in combination with Eq. (32) yields the equation for Node 1

$$(\beta_1 + f_1 \alpha_1) C_1^{i+1} + (\gamma_1 + f_2 \alpha_1) C_2^{i+1} = \delta_1 \quad (47)$$

Determination of Longitudinal Dispersion

Elder (1959) conducted the first modeling study on dispersion in open channels. That author related K_x to the shear stress acting on the channel boundary under the assumption of a logarithmic velocity profile with depth, but not accounting for velocity variations across the stream

$$K_x = C_e y u^* \quad (48)$$

where

$$u^* = \sqrt{gR|S_f|} \quad (49)$$

where y = flow depth; u^* = shear velocity (L/T); C_e = dimensionless coefficient equal to 5.86; R = hydraulic radius (L); and S_f = friction slope. Flume measurements were used to determine C_e . Zerihun et al. (2005b) and García-Navarro et al. (2000) adopted the Elder equation to model solute transport with borders. Zerihun et al. (2005b) produced reasonable predictions relative to their experimental data with constant set to 4, but García-Navarro et al. (2000) concluded that Eq. (48) underpredicted K_x by a factor of 10 under the conditions of their study.

Interest in methods for characterizing K_x is mostly related to the study of dispersion in natural streams (Wallis and Manson 2004), where channel geometry, and therefore velocity gradients and transverse mixing, are complex in comparison with the laboratory conditions of the Elder study. Not surprisingly, studies have shown that Eq. (48) severely underpredicts K_x when applied to river flows, where the measured values can vary by orders of magnitude (Fischer et al. 1979).

Burguete et al. (2009) estimated K_x with the following equation developed from river flow data, which they attributed to Rutherford (1994):

$$K_x = 10 \sqrt{g \cdot WP \cdot A_y |S_f|} = 10 \cdot WP \cdot u^* \quad (50)$$

Eq. (50) yields larger K_x values for the same flow conditions than Eq. (48) because it is a function of wetted perimeter WP instead of flow depth. The measurements of Burguete et al. (2009) were reasonably well predicted with their solute transport model in combination with Eq. (50), but they tested their model only using a narrow border.

In their fertigation model, Perea et al. (2010) calculated K_x with the Fischer equation (Fischer et al. 1979). This equation was initially developed considering the structure of the transverse velocity and mixing coefficient across a stream's cross section. The initial expression was then simplified using empirical relationships developed from dispersion measurements in rivers (Fischer et al. 1979; Wallis and Manson 2004), which yielded

$$K_x = 0.011 \left(\frac{v^2}{u^*}\right) \left(\frac{W^2}{y}\right) \quad (51)$$

where W = flow top width; and other variables are as previously defined. Although Perea et al. (2011) produced accurate predictions in comparison with their solute dispersal measurements with, their tests were confined to furrows. Clearly, the Fischer equation will predict much larger K_x values than Eqs. (48) and (50) for similar flow conditions because of the contribution of the W^2 term. Hence, it seems unlikely that Eq. (51) can be used with border irrigation because the top width is very large in comparison with flow depth in those systems.

Some authors have recognized the limitations of the Fischer equation, even when applied to rivers, and thus have proposed enhancements to that approach. One such improvement was proposed by Deng et al. (2001) based on relationships they developed for the cross-sectional area of rivers and resistance, and measurements of transverse mixing. The equation is

$$K_x = \frac{0.15}{8\varepsilon} \left(\frac{W^{5/3}}{y^{2/3}}\right) \left(\frac{v^2}{u^*}\right) \quad (52)$$

where

$$\varepsilon = 0.145 + \frac{1}{3520} \left(\frac{W}{y}\right)^{1.38} \left(\frac{v}{u^*}\right) \quad (53)$$

All variables have been previously defined. An important difference with respect to Eq. (51) is the power of the top width term. This has a significant impact at the scale of surface irrigation flows. Comparisons of K_x values predicted with Eqs. (48)–(52) will be presented subsequently.

Implementation in Code

The proposed solute transport model was added to WinSRFR as a class (SoluteTransport) of the SRFR dynamic-link library (DLL). SoluteTransport is a subclass of a parent class that was developed to handle different types of constituent transport calculations, to be implemented in the future. In the SRFR DLL, the Irrigation class handles the hydraulic simulation (Bautista et al. 2016). A SoluteTransport object cannot be instantiated independently of an Irrigation object, but the reverse is true. Thus, for a given data set, users can select to run the hydraulic simulation alone or in combination with the ADE model. In the latter case, the methods of SoluteTransport run subsequently to those of Irrigation using the output data objects created by Irrigation. This gives SoluteTransport access to the computational grid and to flow variables for each computational node.

Test Cases

Data sets collected by García-Navarro et al. (2000), Playán and Faci (1997), and Perea et al. (2011) were used in this analysis. They are identified in Table 1 and in the remainder of this article simply as the Garcia, Playan, and Perea data sets. The Garcia data set was collected on a border that was covered with plastic overlaid with gravel to prevent infiltration. Four tests were reported, and the one used here is the Unsteady test 1. Because water depths were not measured, the Manning n was calibrated from the measured final advance time (because infiltration was zero), and the dispersivity K_x was calibrated from a separate test conducted under steady-state conditions. It is interesting that García-Navarro et al. (2000) reported a Manning roughness coefficient of 0.04 for a steady test

Table 1. Data sets used to test the solute transport model

Identifier	Garcia	Playan	Perea
System type	Free-draining border	Blocked sloping border	Free-draining furrow
Geometry	$L = 200$ m $W = 2$ m $S_0 = 6.7 \times 10^{-4}$	$L = 255$ m $W = 3.84$ m $S_0 = 0.001$	$L = 168.5$ m $FS = 1.0$ m $B_0 = 0.07$ m $SS = 2.71$ $S_0 = 1.6 \times 10^{-4}$
Inflow	$Q_{in} = 4.8 \times 10^{-3}$ m ³ /s $t_{co} = 45$ min	$Q_{in} = 17.4 \times 10^{-3}$ m ³ /s $t_{co} = 44$ min	$Q_{in} = 1.4 \times 10^{-3}$ m ³ /s $t_{co} = 235$ min
Infiltration	Impermeable	Kostiakov equation $k = 0.013$ m/s ^a $a = 0.4337$	Modified Kostiakov equation $k = 5.666 \times 10^{-4}$ m/s ^a $a = 0.46$ $b = 2.78 \times 10^{-6}$ m/s
Roughness (Manning n)	0.09	0.03	0.045
Fertigation hydrograph	1 pulse $t_{s0} = 17.2$ min $t_{se} = 20.2$ min $C_0 = 8,220$ g/m ³	1 pulse $t_{s0} = 18$ min $t_{se} = 25$ min $C_0 = 1,535$ g/m ³	6 pulses Starting at 15 min, each pulse lasting 10 min, 15-min intervals $C_0 = 550$ g/m ³

Note: L = length; W = width; S_0 = bottom slope; SS = furrow side slope; B_0 = furrow bottom width; FS = furrow spacing; Q_{in} = inflow rate; t_{co} = cutoff time; k = empirical infiltration constant; a = empirical infiltration exponent; b = empirical infiltration steady rate; t_{s0} = solute injection start time; t_{se} = solute injection stop time; and C_0 = inlet solute concentration.

and 0.09 for the unsteady one, even though the tests were conducted on the same border. Those authors discussed possible reasons for the difference.

The Playan data set also involves a border, but with infiltration. As with the previous test, water depths were not measured; the Manning n was therefore determined from the measured final advance. This implies that n reflects errors in the determination of infiltration. García-Navarro et al. (2000) modeled this scenario with the same K_x value that they determined from their borders with no infiltration.

Perea et al. (2011) conducted four furrow fertigation tests on furrows. The free-draining test identified as Furrow 1 was used here. This test, along with the two described in the previous paragraphs, are particularly useful for evaluating the performance of the ADE model and for quantifying the dispersivity coefficient because they consisted of a solute pulse of short duration relative to the duration of the irrigation event. In contrast with other tests available in the literature, which apply the solute over a large portion of the irrigation, short pulses provide ample opportunity to observe the rising falling legs of the solute hydrographs and to quantify the spreading of the plume.

Results

Fig. 2 presents simulation results computed with the Garcia data set using pure advection. This example presents special challenges because no infiltration takes place and the solute pulse is short-lived. The graph represents the translation of the input pulse as a function of time, with each hydrograph corresponding to the distances indicated in the figure. The time scale is time from the start of the irrigation, and the vertical scale represents the relative concentration, i.e., the ratio of concentration at the indicated station and the inlet concentration. Although no water infiltrates, flow is unsteady and velocities are changing along the furrow, which should lead to some spreading of the pulse, especially after cutoff (which occurs prior to the final advance time). Evidence of numerical dispersion is

provided by the broadening of the base of the pulse at different distances with little attenuation of the peak.

The two sets of results presented in the graph correspond to computations with the Akima and cubic spline (CuSpl) schemes. Whereas the Akima scheme eliminated small oscillations produced by the cubic spline solution, ultimately both methods produced extreme solute relative mass-balance errors, 50% and 56%, respectively. Errors of this magnitude are unusual, but they can happen as a result of the SRFR discretization strategy in combination with particular properties of a test. Thus, it is important to understand the source of the extreme errors of this example.

Fig. 3 helps explain the problem. The figure presents the computational grid generated by the SRFR engine for the scenario and the evolution of the advected pulse in space and time for the example, shown in Fig. 3(a), and for a modified version of the scenario, shown in Fig. 3(b). The adjusted scenario consists again of a 3-min solute pulse, but the injection starts 10 min earlier. In both cases,

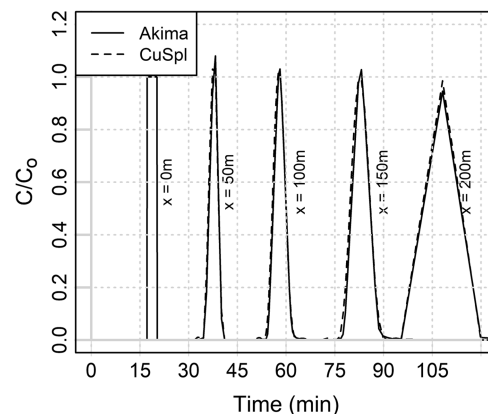


Fig. 2. Pure advection test with two interpolation schemes. Predicted relative concentration as a function of distance and time for the Garcia data set.

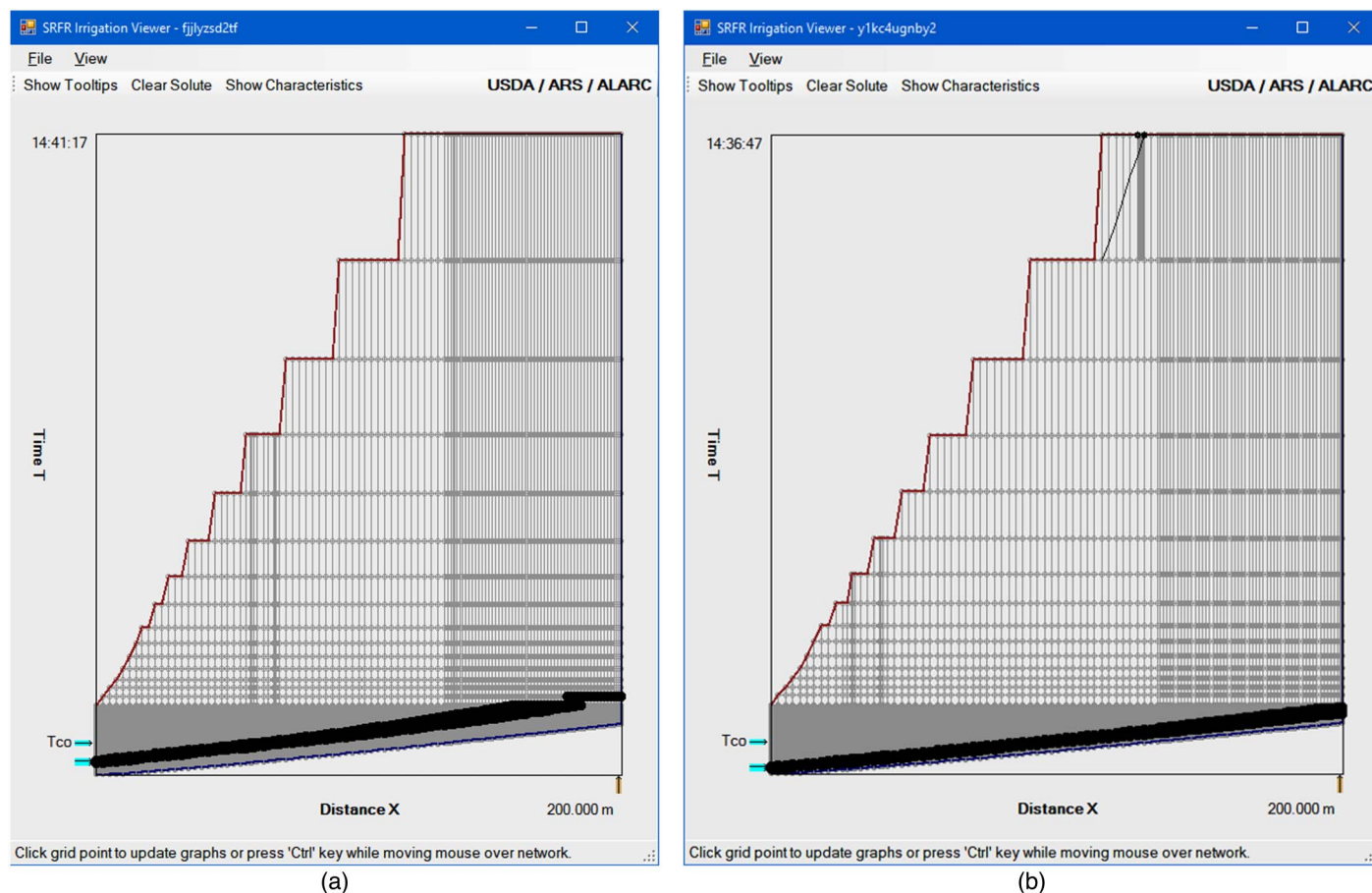


Fig. 3. Computational grid and evolution of the solute pulse in space and time for two fertigation scenarios: (a) pulse arrives to the field end as recession begins; and (b) pulse arrives to the field end before recession begins.

water flow calculations were conducted using the default cell density (i.e., number of spatial increments) offered by the SRFR engine, 80 cells. The images were generated with a SRFR tool, the Network Viewer. Computational cells are very small prior to the start of recession and difficult to discern.

After the start of recession, the time needed to dewater the next remaining wet node determines the time step. Because the scenario involves no infiltration, recession is unusually slow, which in turn makes the recession time steps unusually large. Individual cells are easier to distinguish in the diagrams with the large time steps. The solute plume is the solid area, which in Fig. 3(a) infiltrates completely after the first recession time step. The image shows, therefore, that in the original scenario the solute pulse is reaching the end of the field as the time step of the calculations increases. That problem is not experienced with the scenario in Fig. 3(b), where the pulse reaches the end of the field prior to recession, and the time step remains small as the entire pulse exits the field. Consequently, the absolute mass balance error of the modified scenario is only 2%.

It should be clear that this problem can be mitigated by increasing the cell density because this automatically reduces the time step of the recession calculations. In this case, increasing the cell density to 200 minimizes numerical dispersion and the mass-balance error. However, the relationship between cell density and the error is erratic, and numerical testing is needed to determine a recommended cell density for the given conditions. In this example, the error is minimized with a cell density of 58 and 200, but numerical dispersion effects are evident with the smaller number of cells.

For practical use, a computational strategy for addressing this or similar types of problems that does not require user intervention is needed. This issue is currently under investigation.

Additional pure advection tests have been conducted under different water flow and fertigation scenarios. In general, those results show that numerical dispersion effects and the resulting mass-balance errors are less extreme with typical irrigation scenarios in which water infiltrates along the field and where the solute pulse lasts for a long time relative to typical calculation time steps. The implication is that the SRFR-generated computational grid typically will lead to accurate advection solutions. Both interpolation schemes used here are sufficiently accurate for purposes of these calculations. The cubic spline interpolation scheme tends to produce more noticeable oscillations in the resulting concentration hydrographs, and mass-balance errors typically are slightly larger than when using the Akima scheme. The latter method is easier to implement in code but offers no discernible advantage in computational time.

Fig. 4 depicts the results of tests conducted with the full ADE model and the Garcia data set. The dispersivity was calculated with the Elder equation but with the C_e coefficient adjusted to produce an average K_x equal to the calibrated value reported by García-Navarro et al. (2000), $0.075 \text{ m}^2/\text{s}$, based on flow conditions at cut-off time. The single solute pulse was measured at 50, 100, and 150 m, and those are the results presented in the figure.

In comparison with the measurements, the predicted solute pulses arrived slightly earlier and are slightly more attenuated, especially at $x = 100 \text{ m}$. Similar results were reported by

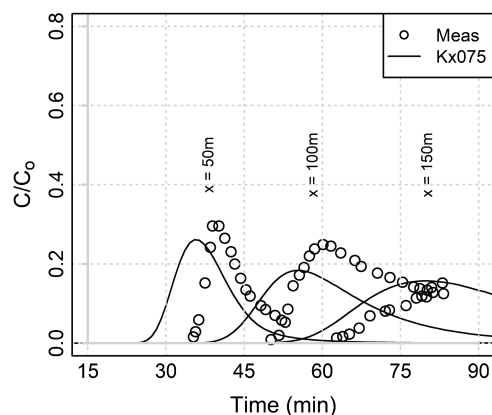


Fig. 4. Advection-dispersion test for the Garcia data set: measured (Meas) and predicted (Kx075) relative solute concentration as a function of time for distances $x = 50$, 100 , and 150 m.

García-Navarro et al. (2000). Thus, these predictions support those made previously, despite some differences in computational methods. In contrast with this model, García-Navarro et al. (2000) represented the water flow with the full hydrodynamic equations. Furthermore, they solved those equations using the McCormack explicit scheme, which requires satisfying the Courant condition when determining the time step of the calculations for a given spatial discretization. In principle, this should lead to a grid that is better suited for the ADE calculations. This shows that the method of tracking segments of characteristics across multiple computational cells, as used here, is effective.

It is important to consider as well that the results of Fig. 4 were calculated using 80 computational cells, as in Fig. 2, but the resulting mass balance error was nearly 0%. This shows that actual dispersion will mask the effects of numerical dispersion and reduce the mass-balance error if the actual effect is sufficiently strong. The practical implication is that although the quality of the advective solution needs to be studied as a function of cell density, the accuracy of the solution is less affected by cell density when dispersion is included.

García-Navarro et al. (2000) conducted limited sensitivity tests for their example to the coefficient of dispersivity. Of interest is to contrast estimates of K_x derived with the available predictive equations with García-Navarro et al. (2000) calibrated value $K_x = 0.075$ and to test the sensitivity of the ADE solution to that range of estimates. The average K_x computed with the Elder, Rutherford, Fischer, and Deng equations, based on flow conditions at cutoff time, were 0.006, 0.423, 0.088, and 0.162, respectively. Fig. 5 illustrates solute hydrographs predicted with the smallest and largest values, i.e., a 70-fold increase in K_x . The relative concentration peaks decreased by about a factor of 5. Clearly, order of magnitude estimates of K_x are enough for practical calculations.

Fig. 6 presents ADE simulation results in comparison with the measurements for the Playan tests. The single solute pulse was measured at five distances, and the results reported here are from the 125- and 225-m stations only. Two simulation results are included in the graph. The series Kx75 represents results computed with the K_x reported by García-Navarro et al. (2000), $0.075 \text{ m}^2/\text{s}$. As may be observed, the time of arrival of the pulse is well predicted, but the attenuation is not. This is contrast with results presented by García-Navarro et al. (2000), which show that both the arrival and peak relative concentration are well predicted. The reasons for the differences are unknown.

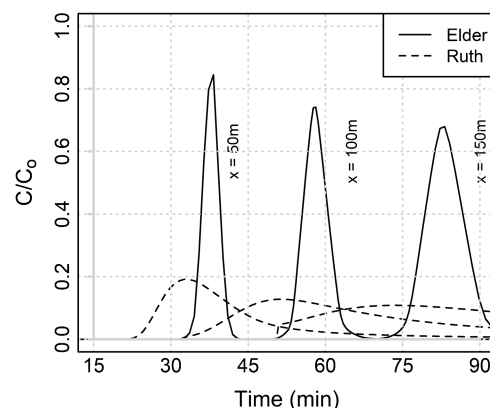


Fig. 5. Advection-dispersion test for the Garcia data set with dispersivity computed with the Elder and Rutherford equations.

Simulations were also conducted with dispersivity estimated with Eqs. (48)–(52). The average K_x (computed based on average flow conditions at the final advance time) were 0.004, 0.72, 3.334, and 0.835 for the Elder, Rutherford, Fischer, and Deng equations, respectively. Thus, these equations provide estimates that vary by nearly 3 orders of magnitude. Of these estimators, the Rutherford equation produced the best agreement with the measurements, and predictions were not very different when K_x was estimated with the Deng equation. The Rutherford results (Ruth) are included in Fig. 6. These results suggest a K_x 10 times greater than for the Garcia data set. The simulation conducted with the Fischer equation, which produced the largest average K_x , produced large artificial oscillations in the computed solute hydrographs. Extreme values of K_x will cause even stronger oscillations, and the ADE calculations will fail.

Of the seven measurements stations reported in the Perea data set, only the 152-m station is illustrated here (Fig. 7). As indicated in Table 1, that test consists of several pulses, which allows comparing the advection and dispersion of those different pulses. The simulation was conducted using the Fischer equation to estimate K_x as was done by Perea et al. (2011). For this test, the average K_x computed with Eqs. (48)–(52) are 0.005, 0.056, 0.008, and $0.043 \text{ m}^2/\text{s}$; thus, values vary by only an order of magnitude, but are smaller than the K_x value reported for the Garcia test.

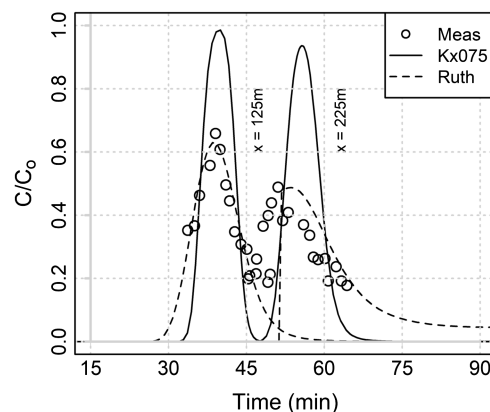


Fig. 6. Advection-dispersion test for the Playan data set: measured (Meas) and predicted (Kx075 and Ruth) solute concentrations as a function of time for distances $x = 125$ and 225 m.

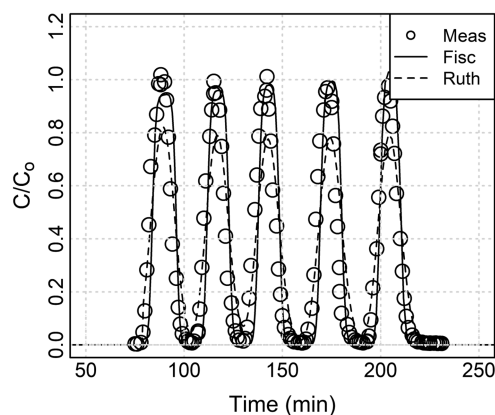


Fig. 7. Advection-dispersion test for the Perea data set: measured (Meas) and predicted (Fisc and Ruth) relative solute concentrations as a function of time at distance $x = 152$ m.

The measurements show very little attenuation of the pulses, which implies that the transport process is advection controlled. Fig. 7 shows predictions generated with the Fischer (Fisc) and Rutherford (Ruth) equations. The former reproduced the measurements with greater accuracy, whereas the latter produced slightly attenuated solute hydrographs.

Discussion

Overall, the proposed solute transport produces results that are consistent with those computed with previously published models and with field measurements. Although the Garcia advection test revealed problems with the advection solution caused by the SRFR discretization scheme, it is important to note again that the conditions of that test are difficult because it involves no infiltration and a solute pulse of very short duration. Under typical irrigation conditions, slight numerical dispersion can be expected from the advection computations, but those effects can also be expected to be eventually overshadowed by actual dispersion, even when computed with the Elder equation ($C_e = 5.86$).

The results presented in the preceding sections offer limited guidance about the determination of the dispersivity coefficient under irrigation conditions. Results suggest larger K_x values for borders (Garcia and Playan tests) than for furrows (Perea test), and perhaps larger values with increasing border width (Garcia versus Playan tests). Excellent simulation results were obtained when applying the Fischer equation to the Perea et al. (2011) furrow data, but that equation seems less useful for borders because K_x varies with the square of the top flow width. An example is the Playan test, for which the K_x computed with the Fischer equation was 1–3 orders of magnitude greater than values computed with other equations, even though the test border was only 3.84 m wide. Much larger K_x estimates can be expected for typical border dimensions of perhaps 50–100 m. The Rutherford equation can likewise produce very large estimates with large top widths.

Additional fertigation data sets were sought to test K_x but the available published data sets (Zerihun et al. 2005b; Brunetti et al. 2018) proved of limited value for that purpose. Those tests applied the solute pulse over at least half of the irrigation duration into blocked irrigation systems. Under those conditions, the flow velocity eventually becomes zero or near-zero, advective-dispersive transport becomes insignificant, and molecular diffusion becomes the primary solute transport mechanism.

Reported fertigation models have proven capable of reproducing measured solute hydrographs with reasonable accuracy under the range of conditions tested by their authors. This model is no exception. This relative accuracy is explained in part by the fact that the fertigation problem is relatively simple in comparison with similar problems such as solute transport in rivers. Irrigation flows are always subcritical and the channel is short and prismatic or nearly prismatic. The solute pulse is well-defined and can be expected to fully mix with the stream by the time the water enters the field. Advection predictions are highly dependent on the calibration of the water flow parameters (infiltration and roughness). Consequently, the travel time of a nonreactive solute plume is well predicted if the water flow is well-predicted. Lastly, because the solute injection typically occurs over a long time relative to the duration of the irrigation event, numerical errors induced by discontinuities in the boundary conditions (start and stop of an irrigation pulse) can be expected to, ultimately, have a limited impact on the accuracy of the solution.

Conclusions

A nonreactive solute transport model based on the advection-dispersion equation was added to WinSRFR, a software package for the hydraulic analysis of surface irrigation systems. Simulations were reasonably accurate under the range of irrigation conditions tested, but results also showed that some inaccuracies could emerge as a result of the discretization scheme used by the hydraulic simulation model, which underlies the ADE solution. Results also revealed limitations of procedures currently available for determining the coefficient of dispersivity. Although solute transport in typical irrigation flows can be expected to be advection-dominated, a better understanding of the range of plausible values for the dispersivity coefficient is needed. Studies need to be conducted under various field conditions to establish this range.

Appendix. Coefficients of the Crank-Nicolson Scheme

The spatial derivative at $k + 1/2$ of Eq. (31) is expressed

$$A_y K_x \frac{\partial C}{\partial x} \Big|_{k+1/2} = \left(\frac{A_y K_x|_{k+1} + A_y K_x|_k}{2} \right) \left(\frac{C_{k+1} - C_k}{x_{k+1} - x_k} \right) \quad (54)$$

or

$$A_y K_x \frac{\partial C}{\partial x} \Big|_{k+1/2} = B_{k+1/2} (C_{k+1} - C_k) \quad (55)$$

with

$$B_{k+1/2} = \frac{A_y K_x|_{k+1} + A_y K_x|_k}{2(x_{k+1} - x_k)} \quad (56)$$

A similar computation for the derivative at $k - 1/2$ leads to

$$A_y K_x \frac{\partial C}{\partial x} \Big|_{k-1/2} = B_{k-1/2} (C_k - C_{k-1}) \quad (57)$$

where the term $B_{k-1/2}$ depends on information from the spatial nodes k and $k-1$.

Applying these expressions to Eq. (31) gives

$$\frac{C_k^{i+1} - C_k^*}{\Delta t_i} = \frac{\theta}{A_{yi}(x_{k+1} - x_{k-1})} \times [B_{k+1/2}(C_{k+1}^{i+1} - C_k^{i+1}) - B_{k-1/2}(C_k^{i+1} - C_{k-1}^{i+1})] + \frac{1 - \theta}{A_{yi}(x_{k+1} - x_{k-1})} \times [B_{k+1/2}(C_{k+1}^* - C_k^*) - B_{k-1/2}(C_k^* - C_{k-1}^*)] \quad (58)$$

and subsequently

$$C_k^{i+1} - C_k^* = \theta \Lambda B_{k+1/2}(C_{k+1}^{i+1} - C_k^{i+1}) - \theta \Lambda B_{k-1/2}(C_k^{i+1} - C_{k-1}^{i+1}) + (1 - \theta) \Lambda B_{k+1/2}(C_{k+1}^* - C_k^*) - (1 - \theta) \Lambda B_{k-1/2}(C_k^* - C_{k-1}^*) \quad (59)$$

where

$$\Lambda_k = \frac{\Delta t}{A_{yk}(x_{k+1} - x_{k-1})} \quad (60)$$

Grouping common terms leaves

$$\alpha_k C_{k-1}^{i+1} + \beta_k C_k^{i+1} + \gamma_k C_{k+1}^{i+1} = \delta_k \quad (61)$$

where

$$\begin{aligned} \alpha_k &= -\theta \Lambda B_{k-1/2} \\ \beta_k &= 1 + \theta \Lambda B_{k+1/2} + \theta \Lambda B_{k-1/2} \\ \gamma_k &= -\theta \Lambda B_{k+1/2} \\ \delta_k &= C_k^* + (1 - \theta) \Lambda B_{k+1/2}(C_{k+1}^* - C_k^*) \\ &\quad + (1 - \theta) \Lambda B_{k-1/2}(C_k^* - C_{k-1}^*) \end{aligned} \quad (62)$$

Data Availability Statement

Some or all data, models, or code that support the findings of this study are available from the corresponding author upon reasonable request. The WinSRFR 5.1 software is available for public download at <https://www.ars.usda.gov/research/software/?modeCode=20-20-05-15>.

Notation

The following symbols are used in this paper:

A_y = flow area;
 A_z = infiltration volume per unit length;
 C = solute concentration;
 g = gravitational acceleration;
 K_x = coefficient of dispersivity;
 L_f = field length;
 Q = discharge;
 R = hydraulic radius;
 S_0 = bottom slope;
 S_f = friction slope;
 t = time;
 t_L = advance time to the end of the field;
 u^* = shear velocity;
 V = flow velocity;
 WP = wetted perimeter;

x = distance;

x_a = advance distance of the irrigation stream;

y = flow depth;

α = coefficient of the Crank-Nicolson finite-difference equation;

β = coefficient of the Crank-Nicolson finite-difference equation;

v = coefficient of the Crank-Nicolson finite-difference equation;

δ = coefficient of the Crank-Nicolson finite-difference equation; and

ε = coefficient of the Deng equation.

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